

Genie is a world model developed by Google DeepMind that generates interactive virtual worlds. It is a spatiotemporal transformer, a type of neural network that models what appears in each frame and how the scene changes over time. First introduced in 2024, Genie was initially trained on video game footage and could only generate two-dimensional environments.

Later versions had higher resolution and increased memory capabilities and could generate three-dimensional environments. Waymo used Genie 3 to create a variant called the Waymo World Model for simulating edge cases for robotaxis. In 2026, Google released Project Genie, a web interface that allows subscribers to use Genie 3. Genie is used in video game design and autonomous vehicles.

Versions

The original Genie model, introduced in March 2024, was only capable of generating two-dimensional environments. It generates frames of the world from interactive input and previous frames, and was trained on video game footage. The original Genie generated worlds at only one frame per second.

Genie 2, introduced in December 2024, had the capability of generating three-dimensional environments as well as two-dimensional ones. It was limited to generating 10–20-second worlds from isometric and first-person perspectives. Compared to Oasis, Genie 2 has greater memory and rendering capabilities. Genie 2 could generate worlds in 360p resolution.

Genie 3, introduced in August 2025, is capable of generating and modifying interactive worlds in real-time. It has a memory of one minute, which is higher than Genie 2's 10–20 seconds. Genie 3 generates interactive worlds at 720p and 24 frames per second, and can produce street environments using Google Street View data.

Technical specifications

Genie is a spatiotemporal transformer that models both the structure within individual frames and the changes between frames. It operates in a latent space, an internal mathematical representation that allows text and image prompts to influence the generated environment. The model contains 11 billion parameters.

Applications

Genie was designed for applications in robotics and simulations. It has been used in video game design, including creating imitations of Nintendo's games Super Mario 64 and The Legend of Zelda: Breath of the Wild.

Self-driving cars

Waymo adopted Genie 3 to create the Waymo World Model, a system for simulating edge cases faced by robotaxis. Compared with Genie 3, it outputs lidar four times faster and produces more realistic simulations. It accepts dashcam or smartphone video as input and provides three forms of simulation control: driving action control, which determines how the model responds to driving input; scene layout control, which specifies the road layout and the behavior of other vehicles; and language control, which specifies environmental conditions. It later helped introduce Waymo's robotaxis to 11 cities in the United States.

Project Genie

Project Genie is a website that allows AI Ultra subscribers to access Genie 3. It was released on January 29, 2026. It is accessible via Google Labs, an experimental products platform. The site limits world exploration to 60 seconds because Genie 3 is an autoregressive model that requires substantial dedicated compute, making longer sessions too expensive to scale to more users. DeepMind says extending the limit would add relatively little testing value while significantly increasing compute costs.

In generated worlds, the WASD keys are used to move, the arrow keys are used to turn the camera, and the space bar is used to ascend. Project Genie has an integration with Google Street View, allowing it to also generate streets. Following its release, the stock prices of several video game companies declined. Thomas Smith of Fast Company reported that Genie could help users create new types of games in the future, games where users can build custom worlds based on their own preferences.

Jay Peters of The Verge found that that it could generate the worlds of Nintendo games but not worlds containing Disney characters, describing its performance as slow. Smith argued that its limitation of 24 fps was insufficient for practical applications, describing it as "too choppy for real-world use", with modern systems capable of achieving 120 fps at the time of its release. TechCrunch reporter Rebecca Bellan also discovered that model generation was limited, noting that it could not generate content related to nudity or copyrighted material.

References

Further reading

Bruce, Jake; Dennis, Michael D.; Edwards, Ashley; et al. (2024). "Genie: Generative Interactive Environments". In Salakhutdinov, Ruslan; Kolter, Zico; Heller, Katherine; Weller, Adrian; Oliver, Nuria; Scarlett, Jonathan; Berkenkamp, Felix (eds.). Proceedings of the 41st International Conference on Machine Learning. International Conference on Machine Learning, 21-27 July 2024, Vienna, Austria. Vol. 235. PMLR. pp. 4603–4623.

Fry, Hannah (August 21, 2025). "Genie 3: An infinite world model. Shlomi Fruchter and Jack Parker-Holder" (Podcast). Google DeepMind.

External links

Official website